

CODE SPORT



In this month's column, we feature a set of computer science interview questions.

Last month, we had discussed a set of questions on operating systems. Let's continue our discussion of computer science questions, while focusing on data management and cloud computing.

1. We are all familiar with transactional databases for storing structured data. However, you are asked to design a data management solution, where the data store needs to store the large documents. The documents are answer sheets of students in a school record management system. Each student is uniquely identified by a student ID, and the data store needs to store all the answer sheets of each student, for all the subjects throughout an academic year, with the ability to store the past three years' records as well. What kind of database would you use? Can you explain the database schema that you would employ?
2. Spatial applications are getting developed in large numbers these days. Consider a taxi fleet management company, where the trajectories of the taxis need to be stored in a database. Spatial applications should be able to query the database for a specific trajectory or segments of a trajectory. It should be possible to answer queries of the form, where given a set of location points, it should be possible to retrieve all the taxis whose trajectories overlap the location points. What kind of database would you use for storing trajectory data? For speeding up query performance, what kind of indexing would you perform for the trajectory data?
3. Assume that you are asked to design the medical data management solution for a big hospital. Medical data generated in the ICU is among the most critical in a hospital, where different sensors generate physiological data also known as vitals. This is a time series data where various physiological signals like heart rate, blood pressure and oxygen level are recorded at periodic time

intervals of, say, one minute - for each patient in the ICU. What kind of database would you use to store time series data?

4. Assume that you are asked to design the data management solution of a small scale social network that is becoming popular. The social network has around 100,000 members and is expected to grow by 10X over the next one year. The members are identified by their unique member IDs. Members can have other members as friends. Each member has a unique profile along with a message history, which includes the messages he/she has sent to other members of the network. What kind of database would you use to represent the members of the social network and their friends?
5. Assume that you are asked to redesign the social network site mentioned in Problem 4 as a dating site. The new constraint on the friendship relationship among the members is that they can only become a friend with a member of the opposite sex. How would you implement the dating site's social network? One of the frequent queries that needs to be optimised for the dating site is: Given two members (X,Y) where X and Y are of the opposite sex, do they have any common connections who can introduce them to each other? How would you design your data management solution such that you can speed up this query?
6. Most of you would be familiar with Dockers (<https://www.docker.com/>), which is an open source project built on top of Linux containers. It allows applications to be isolated from each other. Can you explain how Dockers differs in providing isolation to applications compared to virtual machines?
7. You are given a 16 core, 32GB RAM physical machine running Linux. You need to host a Web